

# SHAHVEZ MEMON

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## PROFESSIONAL SUMMARY

Machine Learning Engineer focused on building end-to-end ML systems — from data ingestion and feature engineering through model training, experiment tracking, and production deployment. Proficient in the Python ML ecosystem (Scikit-learn, XGBoost, Pandas, NumPy) and ML engineering tooling including FastAPI, Docker, MLflow, and Streamlit for serving and monitoring models. Currently applying feature-engineering and modeling pipelines to real production search-ranking data as an ML intern. Seeking Machine Learning Engineer / Data Scientist internship roles, with growing interest in MLOps, deep learning, and applied AI agent systems.

## EXPERIENCE

AI/ML Intern — FlyRank AI

2026 – Present

- Analyze anonymized production search data within an applied search-intelligence track to surface page-level ranking and discoverability signals.
- Build feature-engineering and modeling pipelines — baseline heuristic scoring plus iterative logistic regression, decision tree, and random forest models — validated via client-holdout testing and precision-based ranking metrics.
- Produce explainable, decision-support outputs (ranked review queues, evaluation charts, summary reports) instead of black-box predictions, following team data-safety and reporting standards.

## PROJECTS

Oscar AI Predictor — Production ML Classification System [GitHub ↗](#)

Python · Scikit-learn · FastAPI · Docker · MLflow · Streamlit

- Designed an end-to-end classification pipeline merging IMDb metadata with historical Academy Award records through a modular data-ingestion, transformation, and training architecture.
- Engineered 15+ features including log-transformed budget, gross, and vote counts, ROI, and critic/user review ratios, plus one-hot encoded genre, language, country, and content-rating variables.
- Trained a Random Forest classifier achieving **69.5% accuracy**, **72.6% ROC-AUC**, and **49.0% PR-AUC** on an imbalanced target, with all runs tracked in MLflow.
- Deployed the trained model behind a FastAPI inference service and Streamlit UI, containerized with Docker; added centralized logging and custom exception handling for production reliability.

Calories Burnt Prediction — ML Regression System [GitHub ↗](#)

Python · XGBoost · Scikit-learn · FastAPI · Streamlit

- Built a modular regression pipeline separating data preprocessing, feature engineering, training, and inference into independent, reusable components.
- Trained an XGBoost regressor to predict caloric expenditure, achieving **MAE of 1.22**, **RMSE of 1.75**, and **R<sup>2</sup> of 0.999** on held-out data.
- Deployed the model via a FastAPI REST API with Swagger/ReDoc docs and a health-check endpoint, alongside a Streamlit app for interactive real-time predictions.

Goal-Based Tech Guide Agent [GitHub ↗](#)

Python · Claude API · Prompt Engineering

- Designed a 5-stage agent pipeline that separates deterministic Python intent-detection logic from LLM-based generation, calling the Claude API only after intent and required parameters are resolved.
- Implemented a keyword- and confidence-scoring engine across 9 device and 14 task categories to route requests between direct generation, targeted clarification, and fallback prompts.
- Built structured, decoupled prompt templates and API error handling (rate limits, invalid keys, connectivity) to enforce consistent, reliable step-by-step output.

## EDUCATION

B.S. Computer Science — Sukkur IBA University

Aug 2023 – May 2027 (Expected)

CGPA: 3.30 / 4.00 • Relevant Coursework: Machine Learning, Data Analytics, Database Systems, Statistics, Python Programming

## TECHNICAL SKILLS

**Programming:** Python, Java, SQL

**Machine Learning:** Scikit-learn, XGBoost, Random Forest, Regression, Classification, Feature Engineering, Model Evaluation

**Data Science:** Pandas, NumPy, EDA, Data Cleaning, Visualization (Matplotlib, Seaborn)

**ML Engineering / MLOps:** FastAPI, Docker, MLflow, Model Serialization, Experiment Tracking, Pipeline Architecture, Logging, Exception Handling

**AI / LLM:** Claude API, Prompt Engineering, AI Agents

**Tools:** Git/GitHub, Jupyter Notebook, VS Code

## CERTIFICATIONS & ACHIEVEMENTS

- Kaggle: Python, Intro to Programming, Intro to Machine Learning, Intermediate Machine Learning, Intro to Deep Learning, AI Ethics.
- ProBattle Hackathon (IBA Karachi) — collaborated with a cross-functional team to architect and prototype solutions under strict time constraints.
- Recipient of the STHP Scholarship at Sukkur IBA University for strong academic performance.